North East University Bangladesh

Department of CSE

Course Code: CSE-460

Course Title: Deep Learning Lab

Project Title: Detection of Transactional Fraud in Digital Banking

Submitted To

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Respected Sir,

I am submitting my project proposal titled **“Deep Learning-Based Detection of Transactional Fraud in Digital Banking (Transfer and Cash-Out)”** as part of the course requirements for Deep Learning. The project focuses on detecting fraudulent transactions in digital banking using deep learning techniques, particularly in transfer and cash-out operations.

Introduction:

Financial institutions process millions of digital transactions daily, among which only a small fraction are fraudulent. These frauds commonly occur in digital transfers and cash-out operations, where attackers attempt unauthorized fund movement.

Due to this extreme class imbalance, fraud detection becomes a challenging classification problem, and traditional methods often fail to capture evolving fraud patterns.

This project proposes a deep learning-based system to detect transactional fraud in digital banking, focusing on transfer and cash-out operations.

Objectives:

The main purpose of this project is:

- To build a Deep Learning model capable of detecting fraudulent digital transactions.
- To handle extreme class imbalance in financial datasets.
- To improve fraud detection performance using advanced neural network techniques.
- To prioritize recall in order to reduce financial losses caused by missed fraudulent transactions.
- To analyze and explain model decisions using explainability techniques.

Motivation:

Fraud detection is a critical application in financial cybersecurity and aligns with modern deep learning research trends. The problem involves real-world challenges such as imbalanced data, behavioral pattern recognition, and high-stakes decision-making, making it both academically relevant and practically important.

Expected Outcome

The expected result is a deep learning-based fraud detection system capable of identifying fraudulent transactions with high recall, maintaining a balanced trade-off between false positives and false negatives, and providing a scalable framework suitable for real-world deployment.

Problem & Dataset Overview:

The dataset consists of approximately 6.3 million financial transactions, with only about 0.13% labeled as fraudulent. The task is to build a binary classification model to distinguish between legitimate (0) and fraudulent (1) transactions.

The dataset includes numerical features (transaction amounts, account balances), categorical features (transaction types such as transfer and cash-out), temporal features (time-based information), and behavioral features capturing balance changes before and after transactions. Fraud is mainly concentrated in transfer and cash-out operations.

Methodology:

The proposed system includes data preprocessing and deep learning-based modeling. Preprocessing involves data cleaning, log transformation, feature engineering, encoding categorical variables, scaling numerical features, and handling class imbalance using weighted loss or SMOTE.

Multiple deep learning models will be implemented and evaluated, including Multi-Layer Perceptron (MLP), GRU, Bi-LSTM, and TabNet. Each model will be trained and compared under the same evaluation setup to identify the best-performing architecture for fraud detection on tabular financial data. Training will use the Adam optimizer, weighted binary cross-entropy loss, and early stopping based on validation performance.

Evaluation:

The model will be evaluated using Precision, Recall (primary metric), F1-score, ROC-AUC, PR-AUC, and confusion matrix. Accuracy is not prioritized due to severe class imbalance. Recall is emphasized to reduce missed fraudulent transactions, which have higher financial impact.

Explainability:

Model interpretability will be ensured using SHAP (SHapley Additive Explanations). Feature importance analysis will be used to explain predictions, improving transparency and trust in financial decision-making systems.

Future Work:

Future improvements may include Graph Neural Networks (GNNs) for fraud ring detection, real-time deployment using FastAPI, and concept drift handling for evolving fraud patterns.

I believe this project will provide a strong foundational understanding of deep learning applications in financial fraud detection and fulfill the learning objectives of the course.

I would be happy to provide a demonstration and explain the model, methodology, and results in detail at your convenience.

Best regards,
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